# LAPLACE TRANSFORMS: DC MOTOR RESPONSE TO A RAMP INPUT

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## 1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

**Table 1:** *Learning Outcomes for the Practical Exercise*

| ID | Learning Outcome | Achieved |
|---|---|---|
| 1 | Use the Laplace transform of a ramp input and the motor transfer function to derive the speed response in the time domain. | ☐ |
| 2 | Apply a voltage ramp to the DC motor and compare the measured speed with the predicted response. | ☐ |
| 3 | Explain why the steady lag between the output and the scaled input equals the time constant of a first-order system. | ☐ |

## 2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The first-order model of the motor, $G(s) = K/(\tau s + 1)$, identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares, is used throughout. Use the values of $K$ and $\tau$ identified for your own motor. Where example values are needed, this guide uses the A1.4 example results, $K = 28$ RPM/V and $\tau = 0.15$ s.

## 3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. A voltage ramp is applied to the motor, and the Laplace transform is used to predict the speed response before it is measured. The exercise supports the ACE-CORE learning guide, Laplace Transforms.

### Derive the Ramp Response

The ramp starts from a voltage $u_0$ above the dead-zone, where the motor already runs steadily, and rises at a rate $r$ (V/s). Working with changes from this operating point, the input is a ramp $r\,t$, whose Laplace transform is $r/s^2$. The change in speed is therefore:

$$\Delta\Omega(s) = G(s)\,U(s) = \frac{K}{\tau s + 1} \cdot \frac{r}{s^2}$$

Expanding in partial fractions and using standard transform pairs gives the time response:

$$\Delta\Omega(s) = Kr\,(\frac{1}{s^2} - \frac{\tau}{s} + \frac{\tau^2}{\tau s + 1}) \quad \Rightarrow \quad \Delta\omega(t) = Kr\,(t - \tau + \tau e^{-t/\tau})$$

After a few time constants the exponential term has died away, and the speed rises as a ramp that trails the scaled input $K\,r\,t$ by a constant time of $\tau$.

1. Carry out the partial fraction expansion above by hand and check each coefficient.
2. Using your motor's $K$ and $\tau$, calculate the predicted change in speed after 1 s, 2 s and 3 s for $r$ = 1 V/s.

### Apply the Ramp

**[Screenshot to insert]** *Clock → MATLAB Function (u = u0 + r·(t − 1) for 1 ≤ t ≤ 4 s, else u0 or u_end) → Gain 255/5 → PWM output; encoder speed (RPM) → low-pass filter → Scope. Scopes log u, y and t.*

**Figure 1:** *Simulink Model for the Ramp Test*

**MATLAB Function Block:** generating a voltage ramp from the simulation time

```
function u = fcn(t)
u0 = 1.5;          % starting voltage, above the dead-zone [V]
r  = 1.0;          % ramp rate [V/s]
if t < 1
    u = u0;                        % settle at the operating point
else
    u = min(u0 + r*(t - 1), 4.5);  % ramp for 3 s, then hold
end
```

1. Build the model in Figure 1 and log the voltage, the measured speed and the time as $u$, $y$ and $t$. Use a low-pass filter on the speed, with a time constant of 0.05 s or less.
2. Run the model for 6 s on the hardware.
3. Run the script below to overlay the predicted response on the measured speed.

4. Repeat with $r$ = 0.5 V/s (extending the ramp to 6 s) and compare the time lag with the first test.

**MATLAB Script:** comparing the measured ramp response with the Laplace prediction

```
u_k = u.signals.values(:);  y_k = y.signals.values(:);  t_k =
t.signals.values(:);
K = 28;  tau = 0.15;               % your identified model [RPM/V, s]
r = 1.0;  t0 = 1;  t_end = 4;      % ramp rate [V/s], start and end times [s]

w0 = mean(y_k(t_k > 0.5 & t_k < t0));          % speed at the operating point
tr = t_k(t_k >= t0 & t_k <= t_end) - t0;       % time since the ramp started
w_pred = w0 + K*r*(tr - tau + tau*exp(-tr/tau));   % Laplace prediction
w_ideal = w0 + K*r*tr;                             % scaled input, no lag

plot(t_k, y_k, 'k.', tr+t0, w_pred, 'b', tr+t0, w_ideal, 'r--'); grid on;
xlabel('Time [s]'); ylabel('Motor speed [RPM]');
legend('Measured', 'Laplace prediction', 'K r t (no lag)', 'Location',
'northwest');
```

Example results are shown in Figure 2. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares ($K$ = 28 RPM/V, $\tau$ = 0.15 s), so they show the behaviour to expect rather than measured data. During the ramp, the speed trails the scaled input by 0.15 s, the time constant of the model.

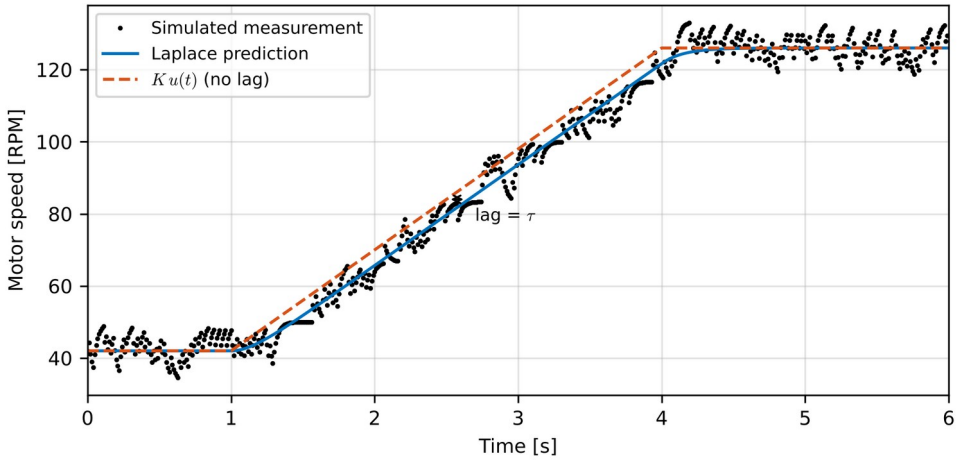



**Figure 2:** *Simulated Example of the Speed Response to a 1 V/s Ramp from 1.5 V to 4.5 V*

**Table 2:** *Ramp Test Results for Your Motor*

| Ramp rate, $r$ (V/s) | Predicted lag (s) | Measured lag (s) | Predicted speed gap (RPM) | Measured speed gap (RPM) |
|---|---|---|---|---|
| 1.0 | | | | |
| 0.5 | | | | |

The speed gap is the vertical distance between the measured speed and $K\,r\,t$ during the steady part of the ramp, and should equal $K\,r\,\tau$. Halving the ramp rate should halve the speed gap but leave the time lag unchanged.

1. Explain, using the partial fraction terms, why the time lag equals $\tau$.
2. How would the prediction change if the ramp started from 0 V instead of from above the dead-zone?
3. Which Laplace transform property turns the derivative in the motor ODE into a multiplication by $s$?

## 4.0 Concluding Remarks

This practical exercise used the Laplace transform to predict the response of a DC motor to a ramp input, and tested the prediction on hardware. Using MATLAB, Simulink and ACE-Lab hardware, the motor transfer function was multiplied by the transform of the ramp, expanded in partial fractions and inverted to give the speed in the time domain.

The measured speed followed the prediction, rising as a ramp that trails the scaled input by one time constant. Working in the s-domain turned the solution of a differential equation into algebra, which is the approach used for all of the transfer function analysis in the remaining exercises.

## Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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